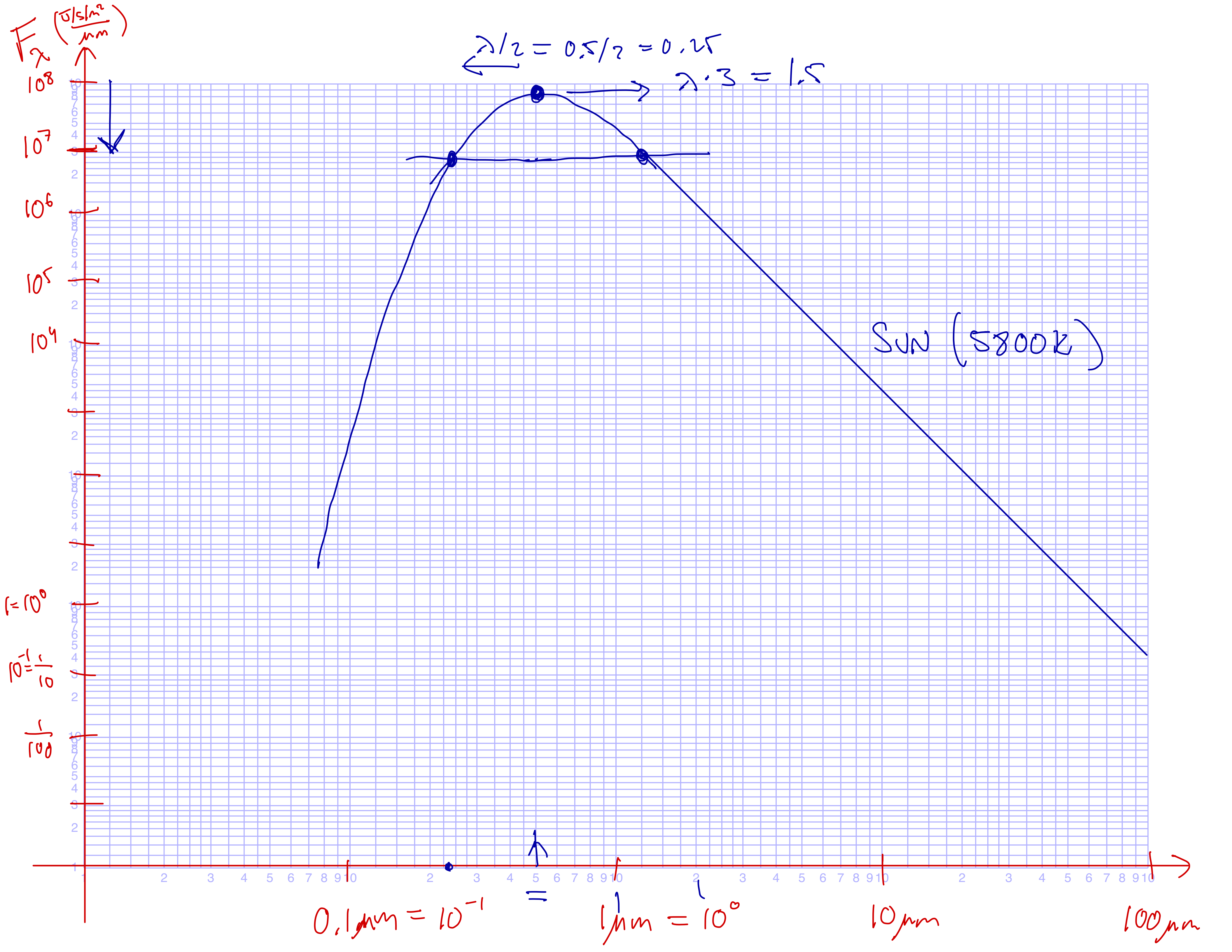
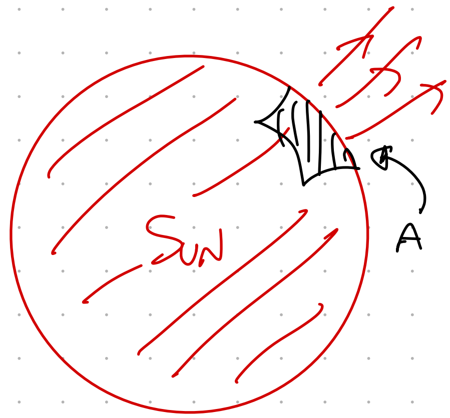


F_λ (J/s/m²/μm)
10^8
10^7
10^6
10^5
10^4
$1 = 10^0$
$10^{-1} = \frac{1}{10}$
$\frac{1}{100}$
$\lambda/2 = 0.5/2 = 0.25$
$\lambda \cdot 3 = 1.5$
SUN (5800K)
$0.1\,\mu m = 10^{-1}$
$1\,\mu m = 10^0$
$10\,\mu m$
$100\,\mu m$

$\boxed{Q2}$

(a)

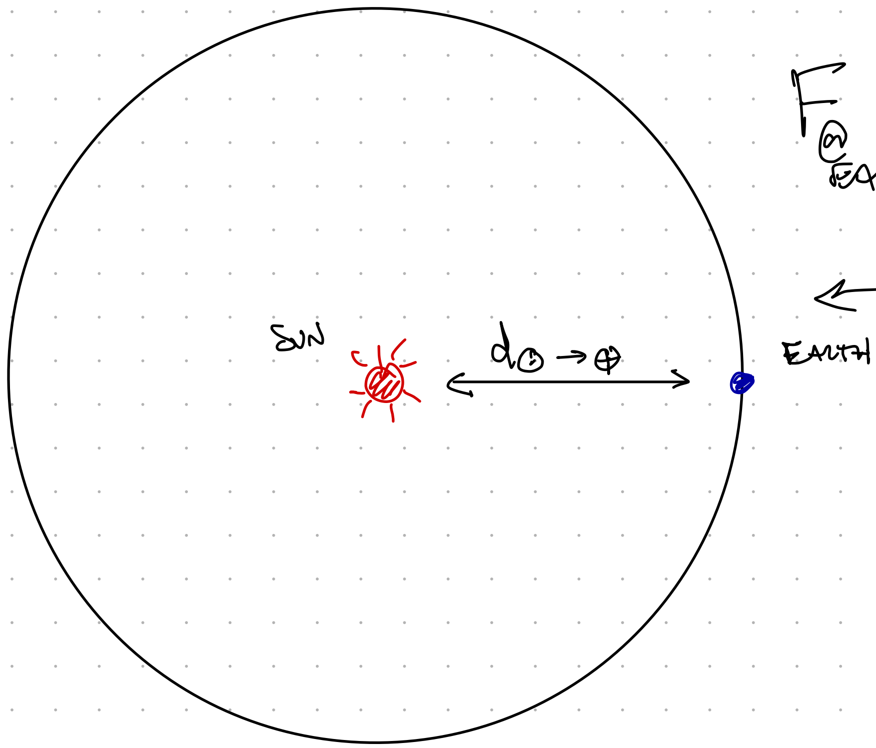
$$F = \frac{J/s}{m^2}$$



$$\frac{E}{\Delta t} = P = F \cdot A \quad \left(\frac{J}{s}\right.$$

$$A_{SPHERE} = 4\pi R^2$$

(b)



$$F_{@\,EARTH} = \frac{P_{@\,SUN}}{\text{AREA OF SPHERE}}$$

$$= \frac{P_{@\,SUN}}{4\pi\left(\frac{d_{S\to E}}{2}\right)^2}$$

(c)